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Feature Review

Continuous downstream processing of biopharmaceuticals

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Continuous manufacturing has been applied in many different industries but has been pursued reluctantly in biotechnology where the batchwise process is still the standard. A shift to continuous operation can improve productivity of a process and substantially reduce the footprint. Continuous operation also allows robust purification of labile biomolecules. A full set of unit operations is available to design continuous downstream processing of biopharmaceuticals. Chromatography, the central unit operation, is most advanced in respect to continuous operation. Here, the problem of 'batch' definition has been solved. This has also paved the way for implementation of continuous downstream processing from a regulatory viewpoint. Economic pressure, flexibility, and parametric release considerations will be the driving force to implement continuous manufacturing strategies in future.

Continuous manufacturing

Continuous manufacturing has been applied in many different industries but not in biotechnology, although bioethanol and microalgae may be the exceptions [1,2]. Continuous manufacturing of recombinant proteins is in its infancy, albeit a lot of companies are currently exploring the economy of continuous processes. One reason for the reluctance to implement these technologies rapidly may be the amount of product on demand in the market: annual market demand may be too low to justify a continuous manufacturing process. Many recombinant proteins are more expensive than noble metals or rare earth elements, and they are produced only in kilogram quantities. In addition, the regulatory framework for producing therapeutic proteins may be prohibitive. Currently, the product is defined by the process; thus, a change from batchwise to continuous processing may require new registration. In the past, the definition of manufacturing batches was debated and was considered the major hurdle for continuous production. Nowadays, health authorities such as the FDA or European Medicines Agency (EMEA) have a clear view on the definition of a batch/production lot in pharmaceutical manufacturing: a batch is a specific quantity of a drug or other material that is intended to have uniform character

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Glossary

Baffles: static devices that regulate the flow of a fluid, in context of mixing they are built in to increase turbulence and/or shear.

Batchwise downstream processing: culture broth is processed in a batchwise manner, a predefined volume is subjected to the individual unit operations to concentrate and purify a product. The feed is usually from batch cultivation.

Camp number (N_{Ca}): a dimensionless engineering parameter used to describe the influence of shear and ageing in precipitation and flocculation.

Centrifual countercurrent chromatographyl: a liquid partition chromatography where a countercurrent motion of the two phases is produced by a complex planetary motion of a coiled tube.

Continuous downstream processing: the culture broth is continuously fed to the individual unit operations required to concentrate, purify, and polish a product. The feed can be from a continuous or a discontinuous/batch cultivation.

Convective chromatography: in convective chromatography the chromatography bed is made of an interconnected network and cast as a single piece either in the form of a membrane or a column. When the bed is cast as a column the bed is also named monolith. In convective chromatography the mass transfer is achieved by convective flow in the pores in contrast to conventional beds packed with porous beads where mass transfer is mainly achieved by pore diffusion. Convective transport is much faster, thus these media have flow-independent mass-transfer characteristics. Resolution and binding does not depend on flow rate.

Downstream processing: all steps required to purify a biological product from cell culture broth to final purified product. This includes cell harvesting, cell lysis if required, refolding if required, initial capture, purification, and polishing of the product.

Flocks: a precipitate that adopts a woolen-like structure during precipitation or flocculation.

McCab–Thiele algorithm: this method uses the equilibrium phase diagram to determine the number of theoretical stages required to achieve a desired degree of separation. It is a simplified method but nonetheless a very useful tool for the understanding of unit operations for separation. The algorithm was originally developed for distillation.

Multicolumn solvent gradient purification process (MCSGP): MCSGP combines two chromatographic separation techniques, which are solvent gradient batch and continuous countercurrent simulated moving bed. The process consists of several chromatographic columns, which are switched in position opposing the flow direction. Some columns are interconnected, so that nonpure product streams are internally, countercurrently recycled.

Periodic countercurrent chromatography (PCCC): in PCCC a column is loaded until full saturation is achieved and the material breaking through is loaded onto a second one. After loading is completed the column is disconnected and material is eluted while loading of the second column is continued and breakthrough loaded on a third column. After regeneration of the first column another cycle can start.

Simulated moving bed (SMB): in SMB a countercurrent movement of the stationary and mobile phase is simulated by a complex interconnection of at least four chromatography columns. It is a quasicontinuous separation method.

Single-pass tangential-flow filtration (SPTFF): a new form of tangential-flow filtration (TFF) without a recirculation loop, providing the performance of TFF with the simplicity of direct-flow filtration. The method can be easily integrated into a continuous process.

Upstream processing: all steps required to produce a compound by biotechnological means, this includes cell line development, production of master and working cell bank, inoculation, small scale cultivation, and manufacturing scale cultivation.

and quality within specified limits, and is produced according to a single manufacturing order during the same cycle of manufacture. A batch refers to the quantity of material and does not specify the mode of manufacture (Code of Federal Regulations: 21 CFR 210.3). This is important for continuous manufacturing, because the development scientists and engineers must define a batch throughout a continuous manufacturing cycle. A batch/lot at the product collection step can be simply defined by the production time period, the production variation such as different lots of feedstock, or by other means. Product can only be collected when the system is in steady state and not disturbed. No specific regulations or guidance for continuous manufacturing are available, other than the definition of 'lot'. So from the regulatory standpoint there is no reason to avoid continuous bioprocessing.

The production of a biosimilar may present an opportunity to change the production process. The process must be relicensed, and there is economic pressure to compete for access to the originator market [3]. Often the original process is outdated and must be revamped [4]. In the past, the sole motivation for developing continuous manufacturing was reduced product stability. Proteins with short half-lives in the culture supernatant or in an intermediate solution must be continuously recovered, otherwise they are progressively denatured. A prime example was the production of recombinant blood coagulation factor VIII; currently, its continuous downstream processing (see Glossary) has been realized at industrial scales [5].

Many of the unit operations for downstream processing of biotechnological products used in batch mode were originally designed for continuous operation. Two typical examples are the continuous disc stack centrifuge and the high-pressure homogenizer for cell lysis. State of the art, high-pressure homogenizers equipped with at least two homogenization valves and allow quantitative cell disruption in one passage.

In this review, the overall scheme of downstream processing of biopharmaceuticals will be elucidated. Moreover, the methods by which continuous upstream processing can be interconnected with continuous downstream processing will be discussed. Finally, details on the individual unit operations and how they can be made continuous will be provided.

Flow schemes and connections between upstream and downstream processes

The entire process for manufacturing of a biotechnological product is divided into upstream and downstream processing. Upstream processing includes the generation of the cell line for overexpression of the protein, establishing the master and working cell bank, inoculum preparation and cultivation in the seed bioreactor, and full-scale cell cultivation. Thus, upstream processing includes more than only cell cultivation/fermentation. Downstream processing encompasses all process steps from cell harvest to the final purified product. The flow scheme for continuous downstream processing does not substantially differ from that for a batch process (Figure 1). Cells are continuously harvested and, depending on the expression system, either clarified culture supernatant or cells are further processed.

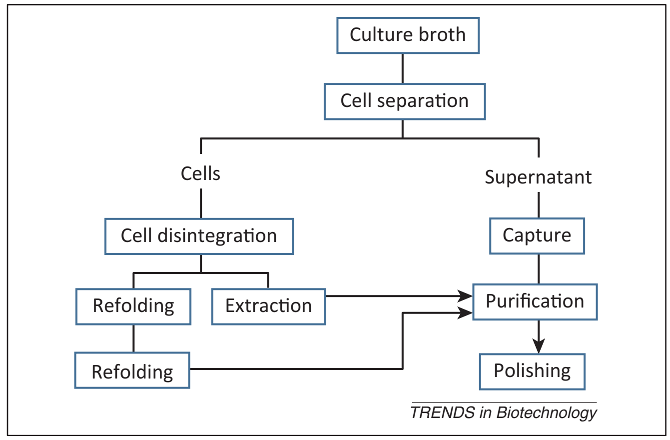

Figure 1. General flowchart of downstream processing of recombinant proteins irrespective of the starting material. Cell culture broth can be bacteria, yeast, fungi, or mammalian cells.

Typically, purification efficiency does not improve with continuous downstream processing; thus, an equivalent number of unit operations must be installed for batch and continuous modes. Exceptions may be found in situations where the product is highly unstable, and instantaneous processing produces a relatively high product:impurity ratio; thus, fewer unit operations would be required in a continuous mode. In general, continuous operations are justified by higher productivity, a smaller footprint, and reduced buffer consumption [6]. In batch and fed-batch fermentation the cells must be expanded to the maximum cell concentration in which cells are still viable. This expansion is only necessary at the beginning of a continuous fermentation process. In continuous operation washing and CIP cycles are reduced and time to reach a steady state is usually negligible compared with the overall process time. For biopharmaceutical purifications, bioreactor harvesting is followed by at least three unit operations, including capture, purification, and polishing.

A robust process requires an intermediate or small holding tank that couples the fermentation effluent with downstream processing. The effluent must be collected under sterile conditions, because the bioreactor should be contaminated through the harvesting pipes. For this purpose, acoustic filters for cell retention [7] were shown to be effective with perfusion cultures for over 50 days [8]. Microfiltration membranes [9], simple settling devices [10], centritech centrifuges [11], and spin filters [12,13] have been used since the very early days of large-scale cell cultures, and these devices have been constantly improved [14]. In addition, cell free culture supernatant can be passed through a sterile filter. Then, a sterile, continuous, cell free feed can be implemented for further downstream processing.

Continuous centrifuges

Continuous centrifuges are readily implemented in continuous processes for recovery of biomass, clarification of cell homogenates, washing and recovering inclusion bodies, or harvesting precipitates or protein crystals. In biotechnology, three common centrifuge configurations are used: (i) tube centrifuge; (ii) chamber centrifuge; and (iii) disc centrifuge (Figure 2). Only the disc stack centrifuge can be

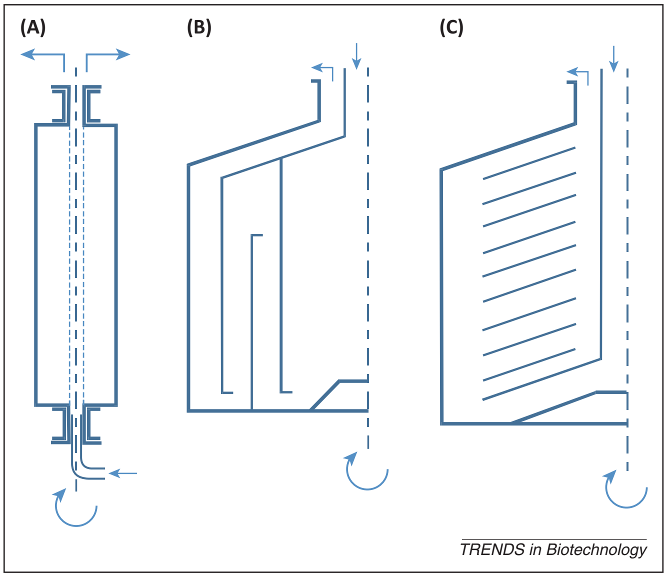

Figure 2. Typical centrifuge configurations used in downstream processing of recombinant proteins. **(A)** Tube centrifuge, **(B)** chamber centrifuge, and **(C)** disc centrifuge. Arrows indicate the direction of solution flow; curved arrows and dashed lines indicate the direction of spin and the spin axis, respectively.

operated in a pseudocontinuous way [15]. Disc stack centrifuges are continuously fed with suspensions, clarified liquid is continuously removed, and solids are intermittently ejected. Two different designs are available: split bowl and disc nozzle. Selection between these two designs depends on the solid content of the feed. For feeds with a high solid content, such as yeast fermentation, split bowl centrifuges are used with a spring pressure or hydraulic pressure from continuous flow of an auxiliary operating liquid; for feeds with low solid content disc nozzle centrifuges are used. Scalability of these centrifuges, particularly scaling down, is important for optimizing continuous processes. The smallest commercially continuous disc stack centrifuge with intermittent solid removal must be fed at a rate of at least 1 l/min. This rate is far too high for laboratory-scale optimization experiments; particularly when a continuous operation must be tested for several hours or days. A scaled-down continuous operation should mimic the shear stress conditions in a centrifuge inlet combined with a conventional laboratory centrifuge [16,17]. Nevertheless, these devices are useful for some general assessments; for example, a certain complex feed stream, like *Pichia pastoris* broth [18], could be clarified with disc centrifuges, although they are not built for preparative purposes. Therefore, a gap remains between small-scale devices and pilot-scale continuous centrifuges. Operated in continuous mode, the pilot centrifuges may be able to serve as full-scale industrial equipment. However, for difficult separation problems, such as clarification of a thick broth, a homogenate, or a fine precipitate, the scaling-up process runs a certain risk, because it may impact the remaining purification steps [19]. There is definitely a need to develop small-scale, continuous centrifuges.

Small-scale decanters, as an alternative to centrifuges, with a throughput of 30 ml/min are commercially available, but the separation efficiency may be too low for many separation problems in the downstream processing of recombinant proteins. The availability of small-scale equipment may be a driver for changing operation modes.

Continuous filtration

Filtration, in the form of microfiltration and ultrafiltration, is a key unit operation in downstream processing of recombinant proteins. Microfiltration is mainly used for cell harvesting. Ultrafiltration is used for concentration of proteins and often in the form of diafiltration for buffer exchange. Ultrafiltration is also used to clear viruses from recombinant DNA produced in mammalian cells. This step is also named nanofiltration, although it is an ultrafiltration with a membrane with a rather low pore size. For cell harvesting filtration is either used in combination with centrifugation or as an alternative unit operation [20]. Moreover, it has been suggested that dynamic filtration systems combined with affinity technologies [21–23] or electrofiltration intensify processes [24]. Operations would save on costs and could streamline production with such a combination. Continuous filtration, also termed cascade filtration, is used in conventional pharmaceutical manufacturing [25]. Continuous filtration is achieved by connecting several filtration units, where the permeate of one filtration unit is fed onto a subsequent unit. The size of each unit may change from step to step.

In the biopharmaceutical industry, continuous filtration may be used to (i) remove solids by microfiltration to (ii) concentrate proteins by ultrafiltration, (iii) perform buffer exchange or conditioning of process solutions by diafiltration, and (iv) act as a virus removal step by nanofiltration (ultrafiltration utilizing small pores). Ultra-scaled-down models of filtration to mimic energy dissipation and flow patterns are available [26,27]. Laboratory equipment is also readily available and can be easily adjusted for continuous filtration. Continuous filtration can be performed in several ways: 'batch topped off', single-pass tangential-flow filtration (SPTFF), diafiltration, and membrane cascades (Figure 3).

Batch topped off

In batch topped off, a membrane filtration unit is equipped with a feed tank, a pump, a membrane filter unit, and control equipment such as a pressure gauge and flow meters. During batchwise operation the retentate is recycled until the desired concentration is reached (Figure 3A). A fed-batch filtration is achieved when feed is added during filtration at the same rate as permeate flow (Figure 3B). This design can accommodate the processing of larger volumes than conventional batch processing. The process solution is frequently recycled and the residence time of the product in the system is much higher. Filter area may be saved at the expense of time. Owing to the longer residence time the method is not suited for labile products.

SPTFF

A real continuous process is single-pass continuous filtration (Figure 3C) or SPTFF. The product must reach the desired concentration by a single pass through the membrane unit. This can be achieved with a cascade of filters. Such a system is tested for IgG concentration where a concentration of up to 225 g/l has been reached [28].

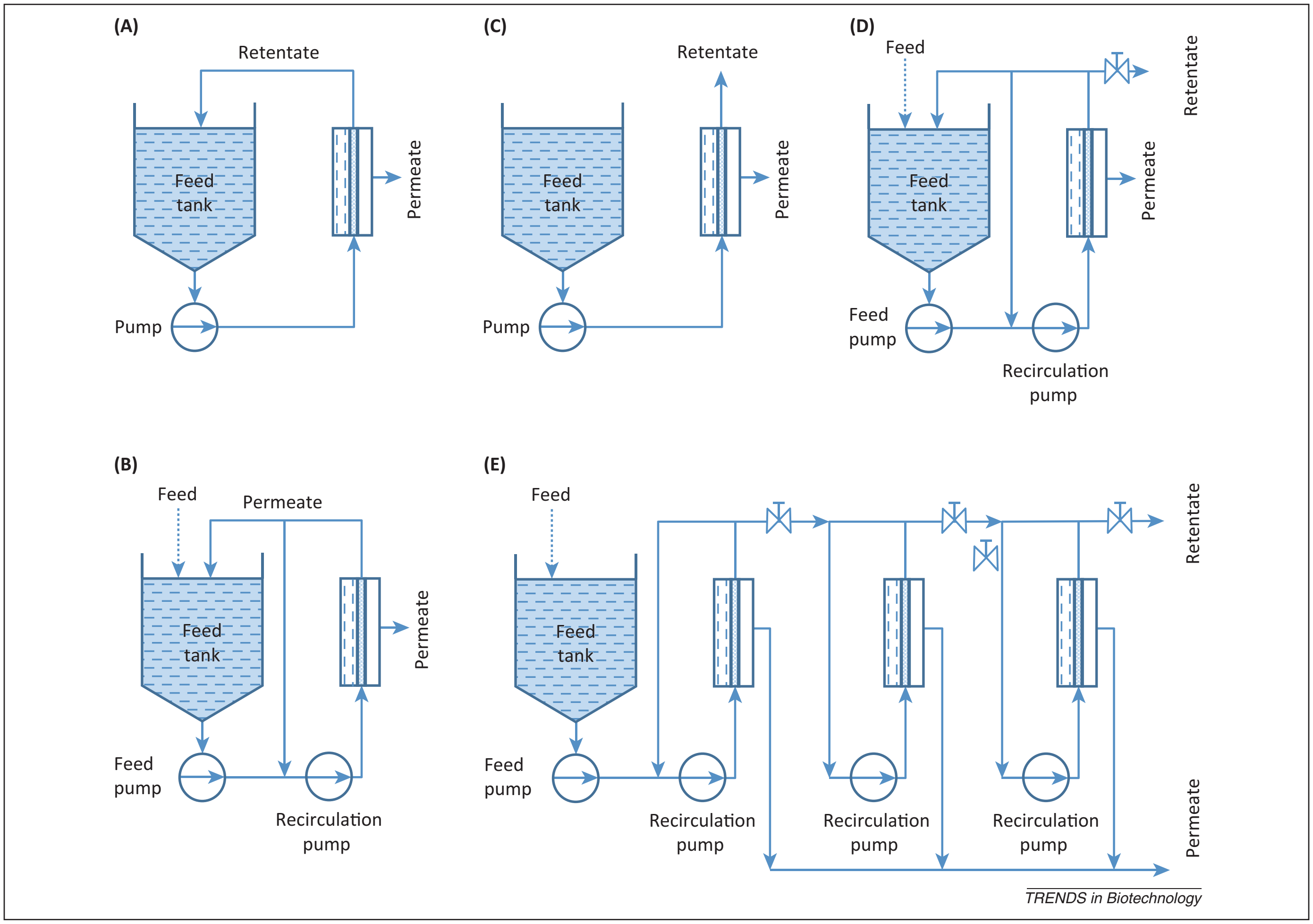

Figure 3. Different designs of batch, fed batch, and continuous membrane filtration. **(A)** Batchwise membrane filtration with recycling of retentate, **(B)** fed batch membrane filtration also named 'batch topped off' with partial recycling of the retentate, **(C)** single-pass tangential-flow filtration, **(D)** continuous membrane filtration with partial recycling of the retentate, **(E)** continuous multistage membrane filtration with sequentially arranged filtration units.

Combination of SPTFF and 'batch topped off' allows processing of large volumes with small feed tanks. The feed tank only serves as a buffer tank when the feed is continuously produced. The partial recycling of the retentate may be required to reach the required concentration. Similar to examples shown in Figure 3C,D, partial recycling of the retentate also increases product residence time in the system with the same consequences as for batchwise operation. The arrangement in Figure 3E also allows continuous filtration. Several membranes can be individually controlled and flux and/or pressure drop according to the concentration (viscosity) adjusted. Although the design is flexible, investment for this system is higher than for SPTFF.

Diafiltration

Diafiltration is the industrial scale unit of operation for buffer exchange and conditioning/washing of cells. The nomenclature of diafiltration in the context of continuous operations is confusing. A discontinuous diafiltration is a membrane filtration process where the solution is diluted with buffer and then concentrated. After several consecutive concentration and replenishing steps the buffer is exchanged. Addition of buffer can be done in a continuous way at the same rate as the permeate flow (Figure 4A). Such a process is called continuous diafiltration, although it is a batchwise operation and the term continuous only refers to the continuous addition of buffers. A real continuous diafiltration can be accomplished by a continuous concurrent (Figure 4B) and countercurrent (Figure 4C) diafiltration. In the concurrent diafiltration process at least three filtration units are connected in series. At the first stage feed and buffer are added at a rate equal to the permeate flux and at all other subsequent stages only buffer is added and permeate is withdrawn. Rate of buffer addition equals permeate flux. The countercurrent diafiltration enables buffer exchange even more efficiently [29]. The fresh feed is continuously added to the final diafiltration stage. In all other stages the permeate is added to the previous stage. In the first stage feed is also added. Countercurrent diafiltration efficiently utilizes buffers and is the most efficient method for buffer exchange.

Membrane cascades

Membrane cascades are another general strategy for downstream processing, separation of proteins beyond concentration, or buffer exchange [30]. Continuous separation of binary mixture by membranes has been realized for mono- and oligo-saccharides [30]. For optimization the McCab–Thiele algorithm originally used for distillation is

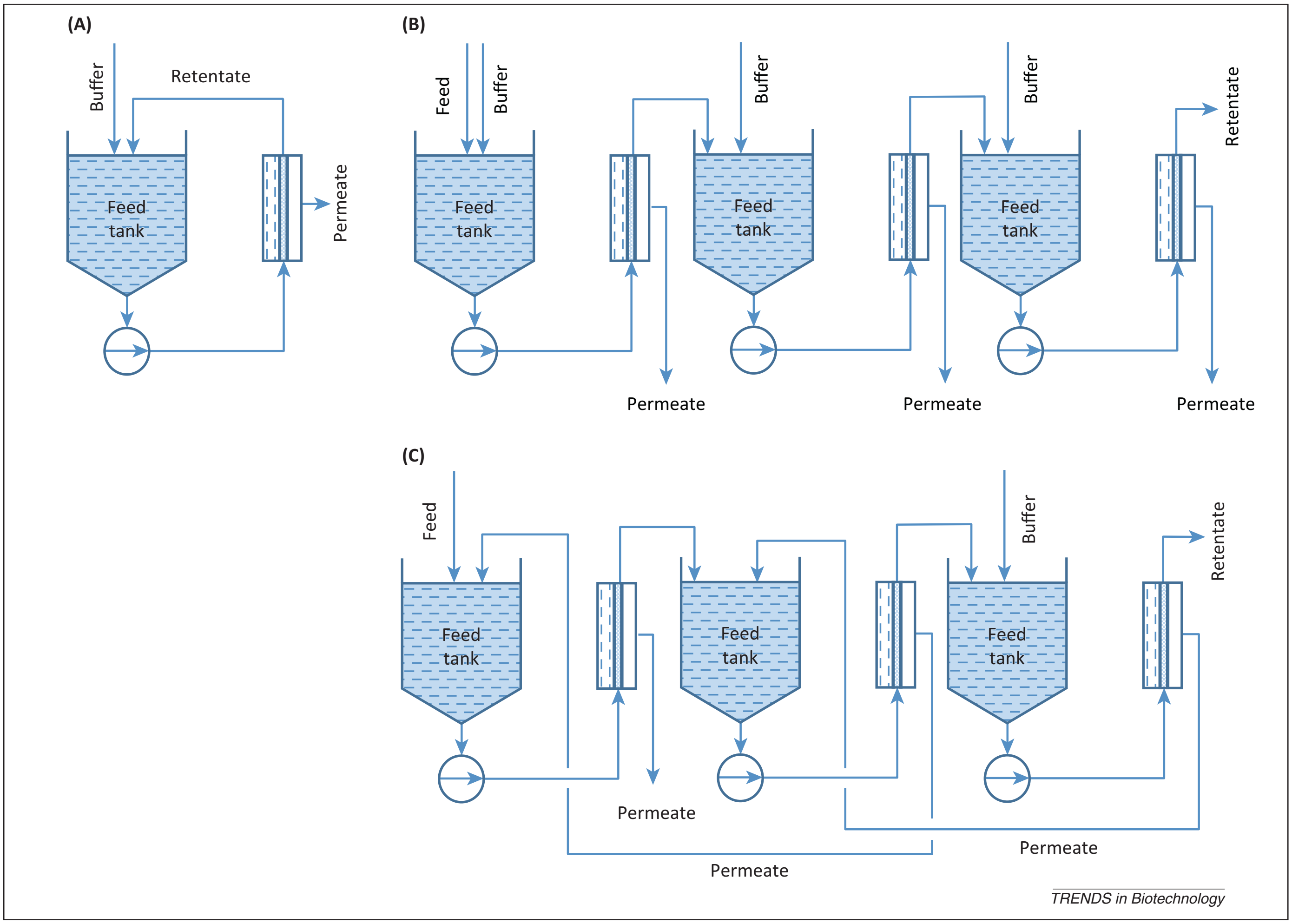

Figure 4. Different modes of diafiltration. **(A)** Continuous diafiltration, **(B)** continuous concurrent diafiltration in multistage mode, **(C)** continuous countercurrent diafiltration in multistage mode.

applied in designing the process. Binary protein separation has been demonstrated with cascade ultrafiltration [31,32]. A continuous cascade filtration in countercurrent mode has been developed for blood plasma fractionation to separate human serum albumin from IgG [32]. When, after the initial filtration stage, a filtration unit is placed at the retentate and another one at the permeate a separation of ternary mixtures can be realized. If the units are designed as SPTFF the process can be continuously operated.

Continuous homogenization and cell lysis

High-pressure homogenizers are the industrial way to lyse cells in pilot and full scale. These machines consist of two main elements: a piston pump usually in the form of a triplex pump and a valve with a slit of around 100 μm to generate pressure up to 1500 bar. Cell lysis is caused by cavitation, which is generated at the valve and immediately behind the valve when the pressure is suddenly released. Modern high-pressure homogenizers can be operated continuously, because the cells are disrupted in a single pass [33,34], which was not possible with older homogenizers [35]. However, when homogenizers are used in continuous mode, cell lysis must be completed in a single pass. This is achieved by installing two homogenization valves in series. Continuous operation is enabled by a triplex pump. The flow rate of modern high-pressure homogenizers is rather high (50 l/h) and unsuitable for laboratory-scale optimization experiments. To mimic the shear stress and flow fields in high-pressure homogenizers small-scale pressure-shear devices are very useful [36]. This type of pressure-shear device can also be used to study biomolecules that resist the high shear rates in high-pressure homogenizers [37]. The effect of shear on DNA disruption during lysis has been intensively studied with these devices. Acoustic cells are more suited for studying the disintegration mechanism, rather than for optimizing a homogenization procedure [38].

Continuous cell lysis has been developed for chemical lysis of bacteria using sodium hydroxide. Several methods based on the same protocol have been described, such as a continuous cell lysis using hollow fiber filters [39] or two mixing chambers to control the alkalinity [40]. A continuous thermal lysis procedure for the large-scale preparation of plasmid DNA has also been developed [41,42]. The harvested *Escherichia coli* cells are treated with lysozyme, heated to 70°C, and then cooled on ice to accelerate lysis. If the product can resist harsh chemical conditions then chemical cell lysis is an option. The method is particularly well suited for plasmid (p)DNA, because the high shear rates must be strictly avoided. High-pressure homogenization would destroy the plasmids.

Continuous precipitation/crystallization

Precipitation and crystallization can be used to capture a product from the culture supernatant (Figure 1). It is a relatively simple method that can be scaled up to the ton scale. The selectivity is lower compared with other methods such as chromatography. Precipitation or crystallization becomes an option when the feedstock has a high titer and fairly high purity; a prime target for this technology is antibodies [43]. The fundamental advantage of continuous precipitation/crystallization is that it provides consistent, reproducible behavior, and it produces particles with optimal size, strength, and density for efficient solid–liquid separations. It minimizes supersaturation and avoids local overshoots and particle break-up. Despite these inherent advantages, continuous precipitation/crystallization is not used for recombinant protein production. An industrial scale ethanol precipitation has been established for blood plasma protein fractionation [44]. Although this type of system is currently available, blood plasma proteins are typically fractionated in batch mode [45].

Mixed suspension and mixed product removal reactors

Continuous precipitation/crystallization has been studied in mixed suspension mixed product removal reactors (MSMPR) [46–48] (Box 1). Over time, the residence time can be controlled, and the growth of flocks or crystals monitored, better than in a batch reactor. For crystallization and precipitation a so-called perikinetic and orthokinetic phase has been defined. In the perikinetic phase the crystals or protein flocks are generated (particle birth) and in the orthokinetic phase the particles/crystals grow to the final size. The growth in the orthokinetic phase is influenced by the power input and thus subject to scale problems. A scaling parameter, the so-called Camp number (N_{Ca}), has been suggested according to Equation 1. This is a combination of shear rate (γ) and time (t).

$$N_{Ca} = \bar{\gamma} \cdot t \qquad [1]$$

γ is defined according to Equation 2:

$$\bar{\gamma} = \left[\frac{P/V}{\rho \upsilon}\right]^{1/2} \qquad [2]$$

where P is the power input in the reactor, V the reactor volume, ρ the density of the liquid, and υ the kinematic viscosity.

Shear stress over time compacts flocks and allows a better separation from the liquid. $N_{Ca} > 10^5$ have been suggested for protein precipitation. Unfortunately this is not the sole engineering parameter for scale up. Precipitates must also be characterized by a fractal dimension, which can be derived from scattering experiments or rheological experiments [49]. The fractal dimension provides some insight into the flock structure. The fractal dimension reflects the orientation of molecules in a system. The more ordered the molecules the higher the fractal dimension. A perfectly ordered system would reach a fractal dimension of three. With increasing power input and time the fractal dimension of precipitate flocks increases.

MSMPR is used more for investigation of precipitation and crystallization rather than for production of proteins. For example, isoelectric precipitation of sunflower protein has been studied in an MSMPR precipitator and was used to model particle size distribution [47]. The ageing characteristics of lysozyme flocks in polyelectrolyte-induced precipitation was established in a MSMPR reactor [48] and stable flocks were reached with N_{Ca} 5×10^5. Aggregation behavior of recombinant proteins was studied in such reactors [28,30].

Tubular reactor

The tubular reactor is an interesting reactor for continuous precipitation of proteins (Box 1). Its advantage is a simple design (Box 1). The protein solution and the precipitant are mixed by a static mixer or even a small stirred tank reactor (STR) and precipitation occurs while the mixture flows through a tube (Box 1). The reactor size is determined by the time required to reach equilibrium and aging of the precipitate. A tubular reactor is often long because a certain shear is required to generate stable flocks and consequently the volume increase due to high flow rate. Baffles or constrictions are also built in to increase shear stress. Precipitation of sunflower protein has also been studied in such a tubular reactor [50]. However, it is clear that, for ageing of the precipitate flocks, time and shear force are important. Ageing just by time will lead to a different flock characteristic than with a combination of time and shear [29]. Thus, the Camp number is not the sole parameter for the design. Higher stability of flocks has been observed in STR than in tubular reactors [51,52] and, as a consequence, a direct change of reactor system without additional experiments is not possible. Protein precipitation has also been investigated with carbon dioxide (CO_2) [32] and transferred to a continuous tubular reactor [53]. It is expected that such reactors might become popular if continuous precipitation is considered an option in industry to intensify manufacturing.

Countercurrent chromatography (CCC) has been further developed for centrifugal precipitation chromatography [54]. Here, a gradient of precipitant is generated in a long separation tube under a centrifugal force field. The protein is repeatedly precipitated and dissolved as it migrates through the separation tube; then, it is harvested in a seal-less centrifuge. Several recombinant proteins, PEGylated proteins, and antibodies were successfully separated [55]. Centrifugal precipitation chromatography is rather complex, and it is not currently used for industrial separations of recombinant proteins; however, it is an interesting alternative to existing processes.

Continuous extraction

Extraction of proteins is also a relatively simple method for purifying and concentrating proteins and has been used a lot for extraction of proteins from natural sources such as plant and animal tissue, blood, and microorganisms. Two immiscible solvents must be found in which the impurities and product differentially partition. Water and organic solvent mixtures are not suitable, because proteins will denature in the organic phase. For proteins aqueous two-phase systems (ATPS) are usually used, because they do

Box 1. Reactors for continuous downstream processing of biopharmaceuticals

Continuous stirred tank reactors (CSTRs)

In reaction engineering a CSTR is defined for homogenous reactions where a stirred tank is continuously fed and product is withdrawn at the same rate. A steady state is reached with conditions depending on reaction rate, feed, and harvest rate. This type of reactor has been extensively evaluated for protein refolding (Figure IA).

Mixed suspension mixed product removal (MSMPR) reactor

Is the equivalent to a CSTR for heterogeneous reactions and is used in studying precipitation and crystallization kinetics. MSMPR reactors have the same performance characteristics, such as exponential wash out kinetics, as CSTRs (Figure IB).

Plug flow reactor (PFR) and tubular reactors

The plug flow reactor is often realized in a tube and thus often used as a synonym. In a plug flow reactor the flow of fluid through the reactor is totally ordered and no portion of the fluid is mixing and with other portions. This reactor type is excellently suited for continuous operations. Several reactions can be continuously mixed at the reactor entrance and then reactions are progressing in the tube while passing through. Virus inactivation by UV radiation is typically performed in tubular reactors. Residence time can be well controlled in this reactor (Figure IC).

Tanks in-series reactor

Connection of at least three CSTR or MSMPR reactors emulates the flow characteristic of a PFR. Thus, this reactor type may be a compromise for scaling up a PFR with sufficient performance. This reactor is suited for fast reactions, for example, addition of precipitant to avoid concentration overshoot or pH adjustment of pH-sensitive protein solutions (Figure ID).

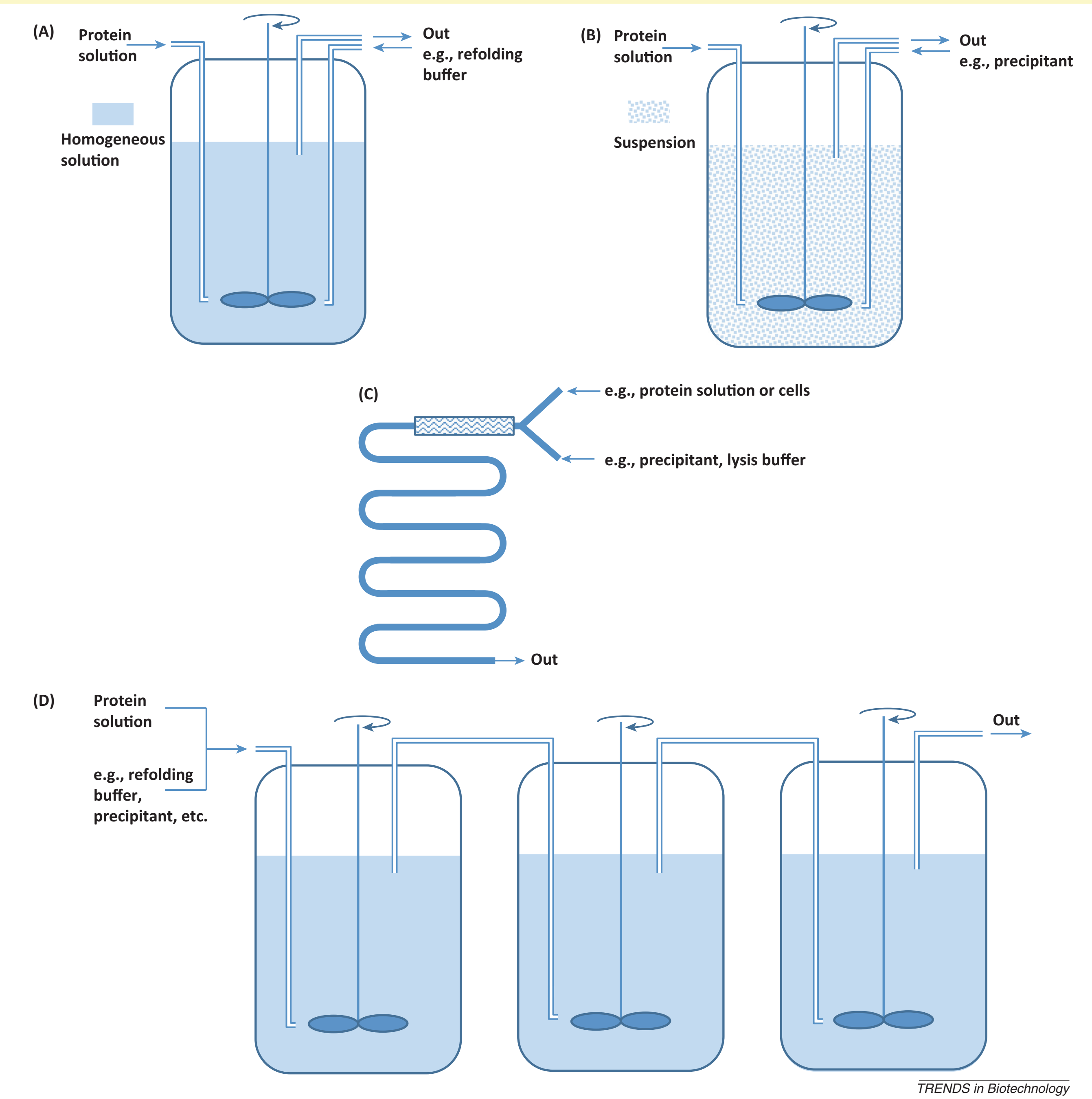

Figure I. Schematic depictions of **(A)** continuous stirred tank reactors (CSTR), **(B)** mixed suspension mixed product removal (MSMPR) reactor, **(C)** plug flow reactor (PFR), and **(D)** tank in series.

not harm protein integrity. Continuous extraction with ATPS was developed in the 1970s. Nowadays, expression systems producing increased titer are available and, thus, the method is also applicable for purification of recombinant proteins [56]. High-throughput methods with microtiter plates, microfluidic devices [57], and molecular modeling have enabled rapid determination of phase diagrams and optimal operating points [58–61]. Antibodies have been the prime application for continuous ATPS extractions. It may be possible to establish a competitive process for purifying antibodies based on ATPS [62], scale up the process [63] and finally convert to a continuous process with a pilot-scale, packed, differential contactor [64]. This process may be economically viable, and its scalability has been demonstrated in the pharmaceutical industry for other classes of products [65,66]. However, current, batchwise antibody purification with conventional, packed bed chromatography has not encountered any limits for scalability [67]. Thus, introduction of novel technologies may be slow, because companies are reluctant to implement a novel process when the existing process is sufficient.

Continuous refolding

Expression of proteins as inclusion bodies allows production of protein at high titers with up to 30 g/l culture broth. Harvesting and initial concentration of the protein is only performed with mechanical unit operations such a centrifugation/microfiltration and high-pressure homogenization. Relatively high product purity, at least 80%, is obtained after inclusion body recovery by differential centrifugation. The drawback is the refolding of proteins, because inclusion bodies can only be dissolved with chaotropic agents and detergents. Protein refolding is a critical step. The *in vitro* refolding process is rather slow; to avoid protein loss due to aggregation refolding must be performed at dilute protein solutions, often less than 100 μg/ml. Consequently, large refolding tanks are required. Thus, refolding adds additional cost in manufacturing recombinant proteins when they are produced as inclusion bodies [68,69]. To overcome these problems three strategies have been applied: adding ingredients to improve solubility or mimic chaperones; segregating aggregates from refolded proteins by performing the refolding in a chromatography column; and combining ultrafiltration with a continuous operation. Continuous refolding would substantially reduce the size of required tanks.

During the past decade, several concepts for continuous refolding have been proposed. Continuous mode provides higher refolding yields than fed batch [70]. For example, one group constructed a paddle reactor that simulated a series of tank reactors in the continuous mode; this method showed higher protein refolding yields than the batch mode [71]. A continuous stirred tank reactor (CSTR) can also be combined with ultrafiltration for continuous protein refolding (Box 1). The refolding buffer and the solubilized inclusion body solution are continuously fed into the CSTR. An ultrafiltration unit is necessary to maintain the levels of redox buffers. For high residence times this continuous operation improves the product yield compared with batch operation. The theoretical model assuming a second-order reaction for aggregate formation and first-order reaction for refolding was experimentally confirmed [72].

Annular and simulated moving bed chromatography

Annular and simulated moving bed (SMB) chromatography have been used for continuous, matrix-assisted, protein refolding. Annular chromatography employs a chromatography matrix packed into a rotating annulus; the eluent buffers and the protein solution are added continuously. In SMB a countercurrent movement of solid and liquid is simulated by periodically switching the inlet ports of the columns. Matrix-assisted refolding employs size-exclusion chromatography (SEC) and ion-exchange chromatography [73–76]. Both methods demonstrated higher yields compared with conventional refolding. In a design where the aggregates are recycled, a theoretical yield of 100% can be achieved, and the folding equilibrium is no longer a limiting factor [74,77]. Thus, higher yields and dramatically reduced buffer volumes can be achieved with these systems. Currently, annular chromatography systems are not commercially available. Similar to annular chromatography, the contact matrix can be combined with a continuous operation with SMB [78,79]. Thus, the refolding efficiency and throughput can be improved simultaneously. Column and operating conditions for different refolding reactions have been explored by Freydel *et al.* [80] with SEC.

Expanded bed

Refolding has also been studied with an expanded bed operated in a continuous mode [81]. This had the advantage of processing proteins in slightly turbid solutions, which are often observed during refolding, particularly at high concentrations of protein or at low concentrations of chaotropic agents. The protein was continuously refolded then captured on an expanded bed.

All continuous refolding procedures have shown high productivity. This was achieved by employing a continuous operation, improving the equilibrium with a contact matrix, or recycling the aggregates.

Continuous chromatography

The simplest way to operate chromatography in a continuous mode is to run several columns in parallel or concurrently [82]. While one column is loaded, the next ones are washed, eluted, regenerated, and finally re-equilibrated. In the simplest form, the number of columns must equal the number of individual steps required to perform the purification. Not all steps require the same time; thus, it may possible to connect two steps in one cycle. Often, loading is the most time-consuming step. Thus, several columns may be idle, and this reduces productivity. To improve productivity, loading may be divided among several columns but this increases the cost of equipment. An industrial scale continuous platform purification process has been achieved with rapid cycling of membrane chromatography units, which can be operated at very high velocity. To date, more than 1 000 000 l of cell culture supernatant have been processed with this platform [5].

High-throughput development and modeling approaches have enabled the rapid development of adsorption and elution conditions at very small scales [59,83–86]. These conditions are typically established in a batchwise mode but can be directly applied to a continuous operation. In principle, one could predict the adsorption equilibrium and kinetics from a single bead; in shallow bed experiments only 5–25 μl of chromatography material is necessary to obtain this information [87,88]. There is no issue with scalability in chromatography applied to biomolecules. Two operation principles have been extensively studied in the area of continuous chromatography: (i) rotating chromatography devices, like annular chromatography (Figure 5B) and carrousel chromatography (Figure 5C; and (ii) CCC and related techniques (Figure 5E–J). The only truly continuous chromatography is annular chromatography. All other designs operate in a pseudocontinuous mode; they are cyclic processes that reach a steady state. Figure 5 gives an overview of the diverse methods for performing chromatography in continuous or pseudocontinuous modes.

Continuous annular chromatography

In continuous annular chromatography (CAC) the separation medium is placed within the annular space of two concentric cylinders. The feed mixture is continuously fed (isocratic operation) on top of the bed covered with a small layer of glass beads or soft gel by a stationary feed nozzle, and the eluent is introduced uniformly to the entire annulus. Additional eluents may also be introduced; the system is not limited to isocratic operation. The column is slowly rotated, whereas the feeding arrangement and the fraction collector are kept stationary. Depending on the adsorption of the components, the eluent velocity, and the rotation rate of the annulus each component will form its own characteristic helical band. The annular chromatography concept has been applied successfully to a number of separations of potential commercial interest.

Protein purification with CAC was pioneered in the early 1990s [89,90]. A scaled-down version [91,92], used for further removal of aggregates [93] and the purification of recombinant green fluorescence protein, has been studied [94]. Here, the same product purity was achieved with continuous mode as with batch mode. Two different separation principles were combined in one column: SEC and ion-exchange chromatography [95]. This allowed integration of two chromatography processes in one and, in addition, a real continuous operation. Further examples of CAC are recovering plasma-derived and recombinant blood

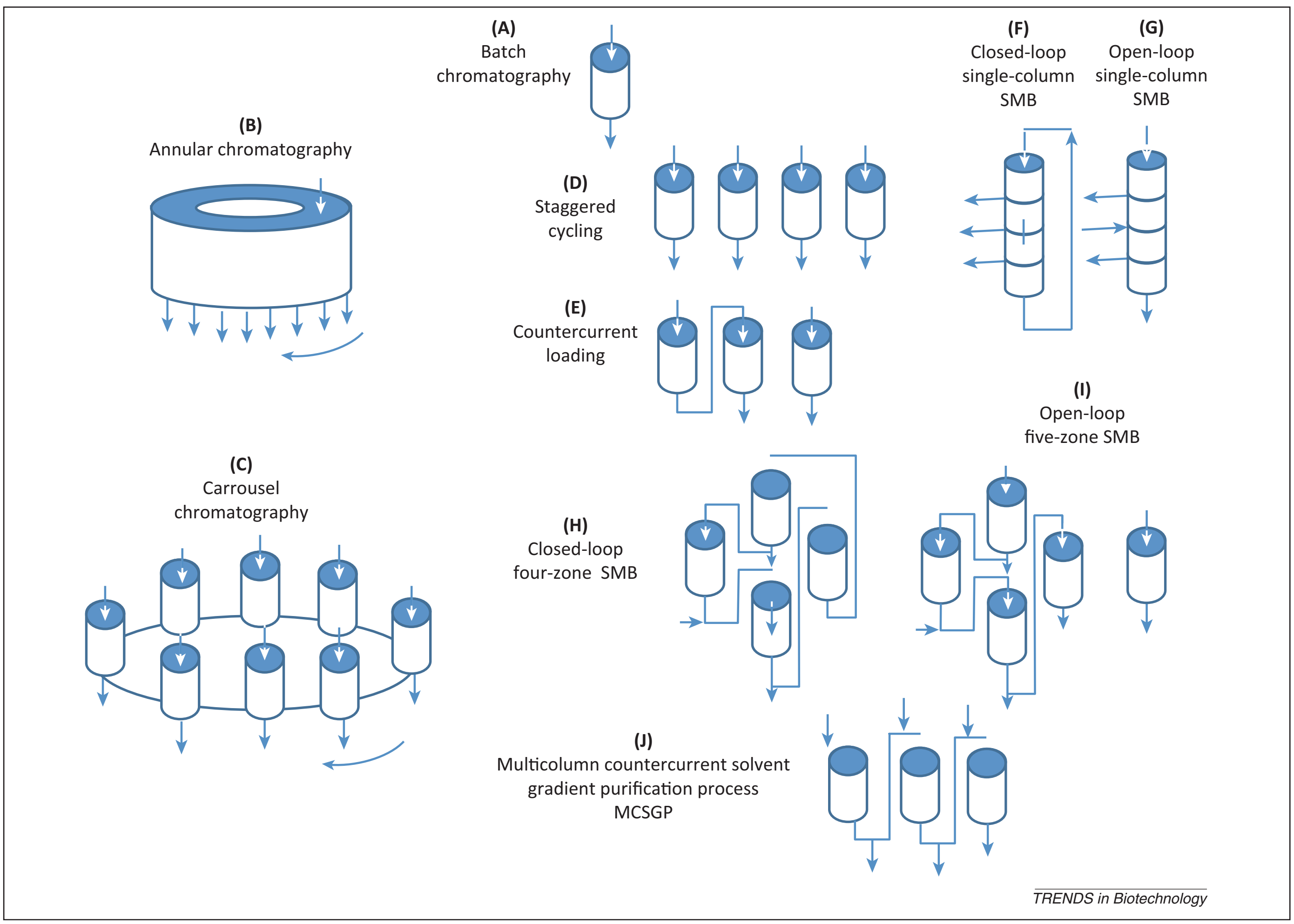

Figure 5. Overview of batch **(A)**, continuous **(B)**, and pseudocontinuous **(C–J)** chromatography processes. The arrows indicate the direction of liquid flow or the revolution of the system in the cases of carrousel and annular chromatography. Only annular chromatography is a truly continuous chromatography, all other configurations operate in cycles and after steady state is reached and a quasicontinuous operation is achieved. For each of the cyclic processes only one cycle is shown. After each cycle the flow direction and feed is changed. All simulated moving bed (SMB) modes **(F–I)** can be operated as closed loop (F,**H**) where the buffer is recycled, or as open loop where the buffer is continuously replenished. Abbreviation: MCSGP, multicolumn solvent gradient purification process.

clotting factors [96,97], capturing antibodies directly from cell cultures [98], and purifying vaccines [99]. Despite many interesting applications with CAC, such as the refolding mentioned earlier in this review, it is not currently used in industry. This method has the potential for substantial process intensification and productivity gain. For example, the rotating mechanical parts require precise machining, and sealing of the bottom plate for a long time period is crucial.

Carrousel chromatography

In carrousel chromatography (Figure 5C) the columns are mounted on the carrousel with interconnecting valves [100]. The columns can be operated in concurrent or in countercurrent mode.

Concurrent and countercurrent loading

The simplest way to operate chromatography in a continuous mode is to run several columns in parallel or concurrently [82]. While one column is loaded the next ones are washed, eluted, regenerated, and finally re-equilibrated (Figure 5D). In the simplest form, the number of columns must equal the number of individual steps required to perform the purification. Not all steps require the same time, thus productivity is smaller with fewer columns.

Productivity can be further improved with countercurrent loading, also known as periodic countercurrent chromatography (PCCC) loading. In the traditional operation of a single column the bed is not fully utilized, because after product breakthrough of unbound material loading is stopped (Figure 5E; Box 2). The nonideal front that forms during loading leaves part of the column unsaturated. Alternatively, the bed can be loaded to saturation point then the column is washed and eluted, similar to a batch operation, while loading the second column is continued [101]. This concept is based on the older, fundamental idea of countercurrent operation [102] and is currently being explored in industry to establish continuous downstream processes [6,103]. The economic benefit is the reduction in chromatography material and buffers and there is an overall reduction in carbon footprint.

PCCC, when operated with two or three columns in series, leads to 15–35% less adsorbent for processes operated in a rapid cycle compared with simple batch operation [104]. Longer cycles lead to smaller improvements in productivity and loading capacity. When the isotherm is flat, larger improvements in loading capacity and productivity can be achieved. This may be possible for proteins directly captured from a culture supernatant with an ion-exchange or mixed-mode resin. Owing to the high levels of salt required, an unfavorable isotherm cannot be avoided [105], but the countercurrent loading strategy may improve the economics. A number greater than two or three columns is not recommended because all subsequent additional increases in yield are very small.

Box 2. Delineation of continuous chromatography for separation of biopharmaceuticals

Cycling
With at least three chromatography columns a quasicontinuous operation is simulated.
Countercurrent loading
At least three columns are required. The first column is loaded to saturation and the breakthrough is loaded onto the second column. The saturated column is washed and eluted with the same methods used in a batch operation.
Simulated moving bed (SMB)
A countercurrent flow between particles and fluids is simulated with a particular arrangement of columns. The original design of the system is suited for the separation of binary mixtures. Variants have been introduced that allow separation of ternary mixtures and application of gradients.
Multicolumn solvent gradient purification process (MCSGP)
The most advanced continuous chromatography process; it requires three columns, it is suited for countercurrent loading, and it offers high-resolution separation.
Annular chromatography
A continuous chromatography method where the chromatography bed is packed into a rotating annulus. The feed solution is continuously applied at the top and fractions are separated at the bottom of the column. This system is suited for multicomponent fractionation.

Multizone SMB

Typically, SMB chromatography can only separate binary mixtures (Table 1). However, binary separation mixtures rarely exist in biotechnology; biotechnological feed stocks typically contain complex impurities that either bind more loosely or more tightly than the product. Thus, various different modifications have been made. Usually SMB consists of four zones to feed and separate the binary mixture and one zone each to regenerate the adsorbent and the eluent (Box 2). A popular modification is a three-zone SMB, with separate columns for elution and regeneration. The columns can be loaded similar to a PCCC; however, in the three-zone SMB the loosely retained impurities are also

Table 1. Overview of continuous chromatography methods for separation of biomolecules

Type of continuous operation	Minimal number of columns	Component	Gradient elution
Parallel operation	3	Multicomponent	Yes
Periodic countercurrent chromatography	3	3	Yes
Three-zone simulated moving bed (SMB)	3	2	No
Three-zone SMB gradient	4	3	Yes
Four-zoneSMB	4	2	No
Four-zone SMB gradient	5	3	Yes
Tandem four-zone SMB	8	4	No
Single column four-zone SMB	1	2	No
Multicolumn solvent gradient purification process (MCSGP)	3	3	Yes
Annular chromatography	1	Multicomponent	Yes

removed in a countercurrent fashion, whereas in PCCC they are not. Thus, the multizone SMB [106] should provide higher productivity than the PCCC.

SMB is ideally suited for SEC [107]. SEC has been used to desalt plasmids [108] and purify viruses [109]. The advantage of operating SEC in SMB mode is that the product is not diluted. By contrast, in conventional SEC the product is diluted five–tenfold. In industrial applications, it is necessary to add an additional column for regeneration with conventional SEC. Biotechnological feed stocks tend to foul the columns and, from time to time, the columns must be stripped. Also, surfactant has been added to SEC for some protein separations [110,111] to improve selectivity and, occasionally, for viral clearance [112].

Tandem SMB has also been proposed for separations of ternary mixtures. In the tandem mode two SMB units are connected, and in the separation process the product is eluted from the first SMB with impurities, but then it is purified in the second SMB [113]. Here, product and migrating impurities are further separated in the second SMB using a different mode of chromatography and thus a ternary or pseudoternary mixture can be separated. A four-zone, single-column SMB has also been used for protein separation [114]. The four zones are realized in one column by dividing the column into zones with built-in flow distributors (Figure 5F,G). Each flow distributor allows the insertion of feed solution or the withdrawal of liquid. This system is capable of compensating for bed shrinkage, which typically occurs in ion-exchange chromatography. This simple design works in a similar way to that of tandem SMB.

Protein chromatography requires a salt gradient for high purity and yield. An alternative system, a multicolumn solvent gradient purification process (MCSGP; Box 2), allows quasicontinuous loading and elution while other steps are performed concurrently [115]. This type of system is not limited to a two-step gradient. Linear, nonlinear, and segmented gradients can be employed. MCSGP is a so-called center-cut separation where the product is recovered from the 'light' and from the 'heavy' impurities. MCSGP also captures the high resolution of gradient chromatography; it can perform challenging purification problems like isoform separation or removal of difficult impurities [116–119]. There are ample potential configurations of chromatography that can be built into a continuous downstream process for producing biopharmaceuticals [120] and offer higher productivity, reduced buffer consumption, and a smaller footprint than current methods.

Increasing chromatography productivity by convective chromatography

Membrane adsorbers and monoliths also facilitate product yield in flow-through chromatography. The product is in the flow through and the impurities are bound to the membrane. This leads to a further increase in productivity for feedstocks, where the impurity is less abundant than the product. Such operations are not only used for polishing but also for high titer feeedstocks, which are produced with defined cultivation media.

Box 3. Outstanding questions

- Continuous downstream processing could be immediately realized with existing unit operations. One major open question regards the gain in economic benefits. This question has not been fully answered and a single answer is not possible, because the answer depends on the production scenario. Such scenarios differ from product to product and companies will tackle this problem in different ways.
- Another pending question is the scale-down of continuous downstream processing equipment. Process development for continuous process requires much more material than a batch process. All engineering parameters for scale-up cannot be estimated from batch experiments. Thus, it is necessary to perform continuous small-scale experiments. In particular long-term operational stability and process control algorithms must be tested at a small scale.
- Ultra-scale-down to process-on-a-chip has limitations, because wall effects and flow pattern are different at this scale compared with laboratory and process scale. Thus, the ultimate limits in scale-down must be explored.

Concluding remarks

The main factors motivating a change in downstream processing from a batchwise to a continuous operation are the gain in productivity, the flexibility, and also the possibility to implement process-control strategies that may be required for a parametric/real-time release. The footprint for continuous operation will be smaller and, often, investment is substantially decreased. Especially in continuous chromatography, the operation becomes more complex but this is outweighed by the gain in productivity and flexibility. The clear open questions regarding economics and scale-down must be answered in order to know which direction continuous downstream processing should be developed (Box 3).

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